

Task 1: AWS Shared Responsibility Model

Task	Responsible Party	Explanation
Data backup	User	The customer is responsible for configuring backups of their data and selecting appropriate backup policies.
User password reset	User	The customer manages IAM users, passwords, MFA and access credentials.
Server patching	Shared	AWS patches the underlying infrastructure, while the customer is responsible for patching the guest OS and applications where applicable.
Network monitoring	User	Customers are responsible for monitoring their cloud resources and configuring appropriate security monitoring.
Encryption key management	Shared	AWS provides encryption services and infrastructure, while customers control how encryption is configured and, in many cases, manage their own keys.

Conclusion:

The AWS Shared Responsibility Model divides security responsibilities between AWS and the customer. AWS secures the underlying cloud infrastructure, while customers are responsible for securing their data, identities, configurations and workloads.

Task 2: Create IAM User zomato-analytics Objective

Create an IAM user with programmatic access and read-only access to Amazon S3.

Steps

Log in to the AWS Management Console.

Search for IAM.

Open IAM → Users.

Select Create users.

Enter:

Username: zomato-analytics

Configure the user for programmatic access using the currently available AWS credential/access-key workflow.

Create the user.

Open the newly created user.

Select Permissions → Add permissions.

Choose Attach policies directly.

Search for:

AmazonS3ReadOnlyAccess

Select the policy.

Click Add permissions.

Verify that the policy appears under the user's permissions.

Expected result:

The zomato-analytics IAM identity can perform permitted read-only S3 operations but cannot modify S3 objects.

Screenshot to attach:

AWS IAM → Users → zomato-analytics → Permissions showing AmazonS3ReadOnlyAccess.

Task 3: Create EC2StartStopPolicy Custom IAM Policy

Create a policy named:

EC2StartStopPolicy

Use the following JSON:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:StartInstances",
        "ec2:StopInstances"
      ],
      "Resource": "*"
    }
  ]
}
```

Steps

Open IAM.

Select Policies.

Click Create policy.

Select the JSON editor.

Paste the policy above.

Select Next.

Enter:

Policy name: EC2StartStopPolicy

Create the policy.

Create the ec2-operator Role

Go to IAM → Roles.

Select Create role.

Select the appropriate trusted entity for your lab scenario.

Attach:

EC2StartStopPolicy

Name the role:

ec2-operator

Create the role.
Expected Result

The role can:

Start EC2 instances.
Stop EC2 instances.

The policy does not grant permissions such as:

TerminateInstances
RebootInstances
DescribeInstances
ModifyInstanceAttribute
CreateSecurityGroup

Note: In a real AWS environment, you may need additional read-only Describe* permissions for some operational workflows or console functionality. The above policy intentionally follows the assignment's strict requirement.

Task 4: Test Unauthorized S3 Access

Objective

Demonstrate what happens when the zomato-analytics user no longer has S3 read permissions.

Steps

Open IAM → Users.
Select zomato-analytics.
Open Permissions.
Remove:
AmazonS3ReadOnlyAccess
Confirm the removal.

Sign in using the IAM user's credentials, or use an appropriate AWS CLI/session configured for that identity.
Attempt to list an S3 bucket.

For example, with AWS CLI:

```
aws s3 ls
```

or:

```
aws s3 ls s3://YOUR-BUCKET-NAME
```

Expected Result

You should receive an AccessDenied or authorization error.

Example:

AccessDenied

Explanation

The operation failed because the IAM user no longer has permission to perform the required S3 action. Under the Shared Responsibility Model, AWS provides the access-control mechanism, while the customer is responsible for correctly configuring IAM permissions and protecting cloud resources.

Screenshot:

Take a screenshot showing the AccessDenied error.

Task 5: Test EC2 Termination Permission

Use IAM Policy Simulator to test the ec2-operator role.

Steps

Open IAM.

Select Policy Simulator if available in your AWS console.

Select the ec2-operator role.

Select the EC2 service.

Select the action:

TerminateInstances

Run the simulation.

Expected Result

Denied

The policy only allows:

ec2:StartInstances

ec2:StopInstances

It does not grant:

ec2:TerminateInstances